Laboratory 06

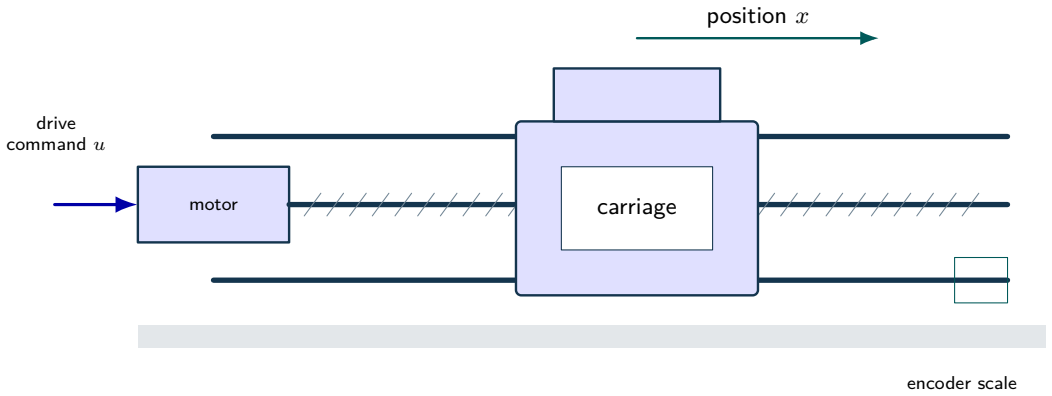
Gain selection with a third-order actuator

Worked solutions – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal unity negative-feedback loop: $G(s) = 1/[s(s+2)(s+6)]$ and scalar controller K. The positioning-axis picture illustrates a possible application; the normalized coefficients are stipulated.

1. A position actuator with an identified aggregate model

The sketch shows a possible motor-driven carriage. Its normalized plant $1/[s(s+2)(s+6)]$ is stipulated as an aggregate teaching model; the poles are not inferred from the drawing dimensions.



Illustrative positioning axis; $G(s) = 1/[s(s+2)(s+6)]$ is stipulated

A motorized positioning axis illustrates the role of the integrator and two stable lags. The normalized transfer function is specified as a teaching model; the pictured dimensions do not identify its coefficients.

(a) Consider unity negative feedback with loop $L(s) = K/[s(s+2)(s+6)]$, $K > 0$. Derive the reference transfer function, characteristic polynomial and open-loop poles and zeros.

Worked reasoning. The reference transfer is $T = L/(1+L) = K/[s(s+2)(s+6)+K]$. Expanding the product gives the characteristic polynomial $p = s^3+8s^2+12s+K$. The open loop has poles at 0, -2 and -6 and no finite zeros. Closed-loop poles depend on K; the open-loop poles are their limiting locations as $K \to 0^+$.

(b) Construct the cubic Routh first column. Find the full strictly stable positive-gain interval and classify its endpoints.

Worked reasoning. The first column is $1, 8, (96-K)/8, K$. All terms are positive exactly when $0 < K < 96$. At $K = 0$, one pole is at the origin, so the system is not asymptotically stable. At $K = 96$, the polynomial factors as $(s+8)(s^2+12)$ and has an imaginary pair. Both endpoints are excluded. Positive polynomial coefficients alone would miss the upper gain restriction.

(c) A numerical step plot appears bounded for $K = 97$ over a short interval. Use the algebraic result to decide whether this is sufficient evidence of stability and specify a decisive numerical check.

Worked reasoning. For $K = 97$, the Routh first column is $1, 8, -1/8, 97$, giving two sign changes and two right-half-plane poles. A short trajectory cannot override this conclusion. Compute all characteristic roots or the state eigenvalues and report their real parts; then extend the simulation with a generic initial perturbation. The duration and excitation can hide slowly growing modes, whereas the polynomial test classifies the nominal linear system.

2. Following branches through a breakaway

Use the open-loop pole map to construct the locus geometrically, then verify the repeated-pole event algebraically. Reject stationary points that belong to the wrong gain sign.

(a) Determine the positive-gain real-axis root-locus segments, asymptote centroid and asymptote angles. Explain how many branches reach infinity.

Worked reasoning. A real point belongs to the positive-gain locus when the number of real poles and zeros to its right is odd. The segments are $(-\infty, -6)$ and $(-2, 0)$. With three poles and no finite zeros, all three branches go to infinity. The centroid is $(0 - 2 - 6)/3 = -8/3$, and angles are $(2q + 1)180°/3$: $60°$, $180°$ and $300°$. The two rightward asymptotes are consistent with loss of stability at high gain.

(b) Find the stationary points of $K(s) = -s(s+2)(s+6)$ on the real axis. Select the positive-gain breakaway point and calculate its gain.

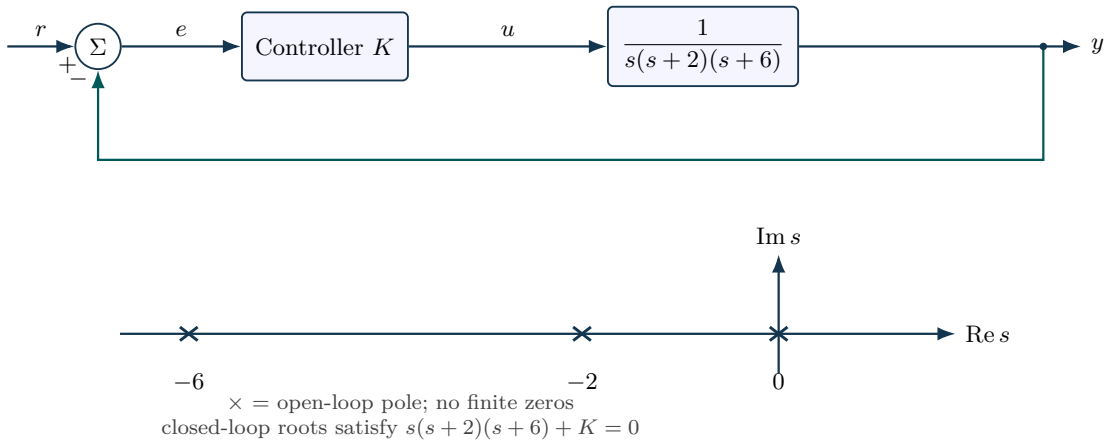
Worked reasoning. Differentiate: $dK/ds = -(3s^2+16s+12) = 0$. The candidates are $s = (-8 \pm 2\sqrt{7})/3$. Only $s_b = (-8 + 2\sqrt{7})/3 \approx -0.902832$ lies on the positive-gain segment $(-2, 0)$. Substituting into $K(s)$ gives $K_b \approx 5.04904$. The other candidate lies between -6 and -2, where positive-gain locus points are absent and its calculated gain is negative. Stationarity alone does not establish a valid breakaway.

(c) At the valid breakaway, determine the third pole without solving another cubic. Check the repeated-pole factorization against the coefficient of s^2.

Worked reasoning. The sum of all three roots is always -8. At the repeated pole $s_b = (-8 + 2\sqrt{7})/3$, the remaining pole is $s_3 = -8 - 2s_b = (-8 - 4\sqrt{7})/3 \simeq -6.19434$. Thus $p(s) = (s - s_b)^2(s - s_3)$ at $K = K_b$. Its s^2 coefficient is $-(2s_b + s_3) = 8$, providing an independent check on branch selection and the repeated-root calculation.

3. Boundary oscillations and feedback-sign mistakes

Classify the nominal loop at both stability boundaries and after reversing the controller sign. Distinguish input–output cancellation from decay of internal states.



Unity negative feedback and the open-loop pole map. Pole symbols provide the starting geometry for the root-locus exercises.

(a) At the upper stability boundary, calculate the oscillation frequency and period and the third pole. Explain whether a bounded-looking oscillation over a finite simulation interval establishes asymptotic stability.

Worked reasoning. At $K = 96$, $p = (s + 8)(s^2 + 12)$. The pair is $\pm j\sqrt{12}$, with angular frequency $2\sqrt{3} \approx 3.46410$ rad/s and period $2\pi/\sqrt{12} = \pi/\sqrt{3} \approx 1.81380$ s. The third pole is -8. The undamped pair does not decay. A finite plot may look bounded even at a marginal boundary or during slowly growing instability, so pole analysis is required for asymptotic stability.

(b) For $K < 0$, prove that at least one real closed-loop pole is positive without using a root solver. Relate the result to the feedback sign.

Worked reasoning. The real polynomial satisfies $p(0) = K < 0$ and $p(s) \to +\infty$ as $s \to +\infty$. Continuity therefore guarantees a positive real root. The closed loop is unstable for every negative K. A sign reversal turns the intended feedback action into reinforcement for the low-frequency positive plant gain. This argument establishes instability without needing the other two poles.

(c) At $K = 0$, the reference-to-output transfer has zero numerator. Does that imply every internal state converges to zero? Explain using the unreduced plant modes and nonzero initial conditions.

Worked reasoning. No. At $K = 0$, the plant retains modes $0, -2, -6$; the zero mode does not decay. The reference is disconnected, so the zero-state transfer from reference may be identically zero while an initial displacement persists. Internal asymptotic stability must be checked in the state dynamics or unreduced characteristic equation. Neither a zero transfer nor a finite plot with no excitation proves decay of all internal modes.

4. A tracking specification with gain uncertainty

Proportional gain must satisfy both a ramp-error specification and strict stability. Treat these inequalities jointly rather than increasing gain until the plot looks acceptable.

(a) For stable K, derive the position and velocity error constants. Find the unit-step and unit-ramp steady errors at $K = 24$. State the stability qualification.

Worked reasoning. The open-loop integrator makes $K_{pos} = \lim_{s\to 0} L(s) = \infty$ and $K_v = \lim_{s\to 0} sL(s) = K/12$. Thus a unit step has zero nominal steady error and a unit ramp has $e_\infty = 12/K$. At $K = 24$, these are zero and 0.5. The final-value reasoning applies because 24 lies inside $(0, 96)$. Applying the same formulas to an unstable closed loop would not yield a valid final error.

(b) Require unit-ramp error no larger than 0.1 using only positive proportional K. Combine the error and stability conditions. What design conclusion follows?

Worked reasoning. The ramp condition $12/K \le 0.1$ requires $K \ge 120$. Strict stability requires $K < 96$, so no gain satisfies both. Even the excluded marginal value $K = 96$ would give the formal ramp error 0.125. The architecture needs modification, such as dynamic compensation, rather than a larger proportional gain. This conclusion concerns the stated linear model and reference class.

(c) Relax the allowed unit-ramp error to 0.15 and select nominal $K = 84$. If actual gain is $84(1 + \delta)$ with $|\delta| \le \varepsilon$, find the largest symmetric uncertainty bound that preserves both requirements.

Worked reasoning. The relaxed tracking condition is $K \ge 12/0.15 = 80$, and stability requires $K < 96$. Therefore $84(1 - \varepsilon) \ge 80$ gives $\varepsilon \le 1/21$, while $84(1 + \varepsilon) < 96$ gives $\varepsilon < 1/7$. The limiting condition is tracking: $\varepsilon_{\max} = 1/21 \simeq 4.7619$ percent. At its lower endpoint, error equals the permitted 0.15; the upper gain is 88 and remains strictly stable. The earlier 0.1 specification remains infeasible.

5. Changing time scale and canceling a stable mode

A redesign changes actuator time scale and adds a compensator. Keep the physical internal mode visible when reducing a transfer function.

(a) Replace the plant denominator by $s(s+2a)(s+6a)$ with $a>0$. Derive the stable gain interval and boundary frequency, and evaluate both at $a=0.5$.

Worked reasoning. The polynomial is $s^3+8as^2+12a^2s+K$. The cubic Routh condition becomes $K<(8a)(12a^2)=96a^3$. At equality it factors as $(s+8a)(s^2+12a^2)$, giving frequency $a\sqrt{12}$. For $a=0.5$, the strict interval is $0<K<12$ and the boundary frequency is $\sqrt{3}$ rad/s. The required gain scales cubically with this time scaling, not linearly.

(b) Suppose the controller is $C=K(s+2)/(s+10)$. Under exact cancellation, derive the reduced characteristic cubic and its stability interval. Explain why the physical realization still contains a stable internal mode and why mismatch matters.

Worked reasoning. Canceling the common stable factor gives loop $K/[s(s+6)(s+10)]$ and reduced polynomial $s^3+16s^2+60s+K$. Hence $0<K<960$. The unreduced characteristic polynomial contains an additional factor $(s+2)$, representing a stable internal mode in a nonminimal realization. Parameter mismatch prevents exact cancellation and changes the transient response. Canceling a stable pole is not automatically destabilizing, but reduced input-output behavior does not describe every internal state.

(c) Combine the changes: use plant $1/[s(s+2a)(s+6a)]$ and controller $K(s+2a)/(s+10a)$. Derive the reduced stability interval, boundary frequency and hidden stable pole; evaluate them for $a=0.5$.

Worked reasoning. Exact cancellation gives $p_{\text{red}}=s^3+16as^2+60a^2s+K$. Routh requires $0<K<960a^3$. At equality, $p_{\text{red}}=(s+16a)(s^2+60a^2)$, so the boundary frequency is $a\sqrt{60}$. The unreduced realization also retains pole $-2a$. At $a=0.5$, the interval is $(0,120)$, the boundary frequency is $\sqrt{15}\simeq 3.87298$ rad/s and the hidden pole is -1. Mismatch breaks the exact cancellation.